



Antibiotic-mediated expression analysis of Shiga toxin 1 and 2 in multi-drug-resistant Shiga toxigenic *Escherichia coli*

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Abstract

Shiga toxin-producing *Escherichia coli* (STEC) is an important foodborne pathogens, known to cause enteric infections especially diarrhea, mainly attributed to Shiga toxins (Stxs). The use of certain antibiotics for treating this infection is controversial, owing to an increased risk for producing Stxs (Stx 1 and Stx 2). Increased antibiotic resistance is also thought to be involved in the pathogenesis of STEC diseases. The purpose of this study was to analyze the effects of antibiotics on induction of Stx 1 and Stx 2 in clinical STEC isolates and to investigate the relationships between increased resistance and Stx production. Fifteen clinical isolates were treated with sub minimum inhibitory concentrations (Sub MIC) of clinically used antibiotics (ciprofloxacin, fosfomycin, tigecycline, and meropenem), and the changes in expression levels of *stx1* and *stx2* genes were estimated using qRT-PCR. The expressions of Shiga toxins were found to be increased up to 6.5- and eightfold under ciprofloxacin and tigecycline Sub MIC, respectively. Fosfomycin had weak induction effect of up to two-fold, whereas meropenem had the weakest influence on such expression. Resistant isolates were found to be more prone to increased expression of toxins.

Introduction

Shiga toxin-producing *Escherichia coli* (STEC) and enteropathogenic *E. coli* (EPEC) strains are responsible for foodborne infections in humans. STEC is a Gram-negative, rod-shaped, facultative anaerobic bacterium which is known to be an important foodborne enteropathogen (Selim et al. 2013). The majority of illnesses due to STEC is reported to be sporadic, sometimes leading to serious outbreaks worldwide (Vally et al. 2012). The morbidity and mortality rates related to pathogenic outbreak of *E. coli* O157:H7 in the USA and a hemolytic uremic syndrome (HUS) outbreak in Germany in 2011 highlighted the potential threats to public health by this pathogen (Basu and Tumer 2015). A joint report of Food and Agriculture Organization (FAO) and

World Health Organization (WHO) revealed that approximately one million annual infections might be attributed to Shiga toxigenic *E. coli* (FAO/WHO 2019).

A greater number of STEC serotypes have been reportedly related to diarrheal infections, mostly isolated from various common animal sources like cattle, sheep, and goats. (Farrokh et al. 2013). Cattle and sheep are the major reservoirs of STEC with fecal contamination of food as the main transmission route (Gyles 2007). However, primary source of *E. coli* in most cases is considered to be the host's own intestinal flora, where it is not pathogenic all the times but can evolve into one through intrinsic or acquired plasmids, transposons, bacteriophages, or pathogenicity islands. The major environmental route of transmission is therefore oral-fecal route through consumption of contaminated food or water (Doughari et al. 2011).

The primary virulence factor of shiga toxigenic *E. coli* is production of the Shiga toxin, which is encoded by a lambdoid bacteriophage, which halts host cell protein synthesis to establish the infection (O'Brien et al. 1984). Along with two major types of Shiga toxins, namely, Stx1 and Stx2, three subtypes of the *stx1* gene (*stx1a*, *stx1c*, and *stx1d*) and seven subtypes of *stx2* gene (*stx2a*, *stx2b*, *stx2c*, *stx2d*, *stx2e*, *stx2f*, and *stx2g*) have been reported (Melton-Celsa 2014). Specific *stx* subtypes such as *stx2a*, *stx2c*, and *stx2d* have

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been reported to be involved in causing human infections and have often been isolated from patients with hemolytic uremic syndrome (HUS) (Friedrich et al. 2003).

Shiga toxins belong to the AB5 family of protein toxins having an A subunit of 32 kDa (A1 27.5 kDa and A2 4.5 kDa) involved in the enzymatic activity of the toxin, and 5 copies of 8 kDa B subunit (B pentamer) responsible for binding to cellular receptors (Basu and Tumer 2015). The target of B pentamer are the cells exposing lipid receptor glycolipid globotriaosylceramide (Gb3) that is the main route for delivering A subunit to the cytoplasm (Fuller et al. 2011). A subunit then targets an adenine within the 28S rRNA of the 60S eukaryotic ribosome subunit, and halts protein synthesis in target cells. The cleavage of this adenine base leads to activation of apoptosis through several stress response pathways (Smith et al. 2003).

Because of its great clinical significance, *E. coli* has been the matter of numerous investigations in order to determine its key virulence factors responsible for initiation and establishment of infections (Cavaliere et al. 1984). The lack of an effective treatment strategy for shiga toxigenic infections has established these as a prominent public health concern. There is also an increasing threat of long-term disability caused by frequent early childhood enteric infections.

The currently recommended treatment of STEC infection is mainly based on supportive therapy and hydration (Thorpe 2004). The use of antibiotics has been associated with worsening of symptoms leading to increased risk of hemolytic uremic syndrome making it controversial (Guerant et al. 2001). Furthermore, antibiotics may intensify shiga toxigenic infections by killing bacteria and consequently increasing the release of shiga toxins that might have been stored in the bacterial cell. In addition, the increased release of Stxs might also be related to the activation of a DNA repair system in *E. coli* known as the SOS response, which reduces the efficiency of treatment (Kimmitt et al. 2000). The SOS response system is made up of more than 50 genes. Antimicrobials are thought to augment the risk of HUS by enhancing the release of Shiga toxins from bacterial cells via a number of ways. DNA damage that can be caused by some antimicrobials may trigger the bacterial SOS response in STEC cells.

The empirical therapy with broad-spectrum antibiotics or drug resistance-induced treatment failures may facilitate emergence and spread of resistance among enteric bacterial pathogens and consequently to human populations. The effects of antibiotics on Stx production in STEC isolates are poorly understood, hence warranting such investigations for increased antibiotic induced production of Shiga toxins in Stx-virulent pathogens. The objectives of this study were to identify the most prevalent types of Shiga toxin in local clinical STEC isolates and to determine antibiotic resistance profiles of each clinical STEC isolates. A panel of clinically

recommended antibiotics was used to investigate their effects on up-regulation of Stx 1 and Stx 2 under Sub MIC of these antibiotics and identify any possible relationship between antibiotic resistance and pathogenesis.

Material and methods

Bacterial isolates

Shiga toxigenic bacterial isolates ($n = 15$) studied for antibiotic-mediated induction of Stx 1 and Stx 2 were collected from Railway General Hospital, Rawalpindi Pakistan from Sep 2016 to July 2017. An additional *E. coli* O157 strain (EDL933) was used as a positive control. STEC clinical isolates were subcultured on Sorbitol MacConkey (SMAC) agar plates. SMAC medium permitted ready recognition of Shiga Toxigenic *E. coli*. The plates were incubated for 36–48 h at 37 °C, and stock cultures were preserved at 80 °C using glycerol 80% (v/v) as the cryoprotectant. Necessary ethical approval for carrying out the study was obtained from Institutional Review Board (IRB), Atta-ur-Rehman School of Applied Biosciences (ASAB), National University of Science and Technology, Islamabad Pakistan.

Antibiotics and antimicrobial susceptibility testing

Antibiotics used for antimicrobial susceptibility testing were purchased from commercial suppliers (Musaji Adam & Sons, Pakistan). Kirby-Bauer Disk diffusion method of susceptibility testing against selected antimicrobial agents (Table 1) was performed on all isolates using the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines to obtain an antibiogram of the MDR isolates (EUCAST 2016).

Detection of *stx1* and *stx2*

Five to seven colonies were randomly selected from each pure culture plate and cultured in 0.8 mL of Luria–Bertani (LB) broth for 24 h at 37 °C with shaking. DNA was isolated by using bacterial DNA extraction kit (GF-1, Vivantis, Malaysia). Samples (4 µL) of each extract were amplified in 20 µL reaction mixtures containing 0.5 µL of DNTPs (0.2 mM), 2 µL buffer (10X) approximately, 1 µL of each primer (1 mM), 0.3 µL of Taq polymerase, and nuclease free water to adjust the total volume to 20 µL. The primers used for *stx1* (171 bp) were as follows: 5' CATTACAGACTATTTTCATCAGGAGGTA and 5' TCG TTCAACAATAAGCCGTAGATTA; and for *stx2* (231 bp);

Table 1 Antibiotic disks and their concentration used for KB disk diffusion antibiotic susceptibility assay

Antibiotic	Concentration (µg)
Penicillin	
Augmentin	30
Amoxicillin	30
Fluroquinolones	
Moxifloxacin	30
Ciprofloxacin	30
Carbapenem	
Imipenem	30
Meropenem	30
Cephalosporins	
Cefepime	30
Cefoxitin	30
Ceftriaxone	30
Ceftazidime	30
Cefazoline	30
Tetracycline	
Tigecycline	30
Aminoglycoside	
Gentamicin	30
Macrolides	
Azithromycin	30
Miscellaneous	
Trimethoprim	30
Chlorempenicol	30
Rifaximin	30
Fosfomycin	30
Colistin	30

5'GCGGTTTTATTTGCATTAGC and 5' TCCCGTCAA CCTTCACTGTA. The PCR reaction mixtures were detected on 1.5% agarose gel, stained with ethidium bromide and visualized under UV transilluminator (Thermo Scientific) at 300 nm.

Minimum inhibitory concentrations of antibiotics using broth microdilution

The minimum inhibitory concentrations (MICs) of four clinically used antibiotics (Ciprofloxacin, Tigecycline, Fosfomycin, and Meropenem) against STEC were determined using broth dilution method (Wiegand et al. 2008) according to the guidelines of the European Committee on Antimicrobial Susceptibility Testing (EUCAST 2016). For fosfomycin, MIC was determined using the agar dilution method on supplemented Mueller–Hinton agar. The MICs determined against each antibiotic were used to calculate sub inhibitory concentrations as 1/2 MIC.

Sub MIC assay

MICs of all antimicrobials were determined for each of the strains. Each strain of Shiga toxigenic *E. coli* was inoculated in 1 mL of Luria broth (LB) and allowed to grow at 37 °C overnight with shaking at 100 rpm. Ten microliters of overnight culture (OD adjusted to 0.05 at A600) was then inoculated into 1 mL of fresh LB medium. The strains were allowed to grow for 2 h, after which the specific antimicrobial was added to the culture at a sub-inhibitory concentration (1/2 of the previously determined MIC). To simulate the actual treatment of antimicrobial agents, we added each antibiotic to broth at 2 h of culture and continued for the overnight incubation.

RNA extraction and reverse transcription

The bacterial overnight cultures grown in 1/2 MICs of the antibiotics were collected, and the total RNA was extracted using Trizol method as explained by Rio et al. (2010). The purity and concentration of RNA were assessed by NanoDrop. The extracted RNA samples were stored at –70 °C until use. The RT-PCR reaction mixture contained 4 µL of 5× reaction buffer, 1 µL of RiboLock RNase inhibitor (Thermo Scientific) (20 U/µL), 2 µL 10 mM dNTP mix, 1 µL of RevertAid reverse transcriptase (Thermo Scientific) (200 U/µL), 500 ng of total RNA, and RNase-free water to make up the final volume to 12 µL. The reaction was carried out at 25 °C for 5 min, 42 °C for 60 min, and 70 °C for 5 min. The amplified cDNA samples were stored at –70 °C until use.

Real time/quantitative PCR

For q-PCR, the Applied Biosystems 7300 Real Time platforms (MyiQ real-time PCR, BIO-RAD) and Power SYBR green Master mixes (Thermo Scientific) were used. cDNAs were quantified to 100 ng/µL in each sample before analysis. Twenty-five-microliter reactions prepared for each processed sample included 12.5 µL of Power SYBR Master Mix, 1 µL each of forward and reverse primer, 1 µL of cDNA, and 9.5 µL of nuclease free water. The 16S rRNA universal primers were used as endogenous internal control for each q-PCR run. There were two technical replicates quantified for each sample. The method used for qPCR data analysis was the comparative CT method (also known as the $2^{-\Delta\Delta CT}$ method) (Livak and Schmittgen 2001).

$$\text{Fold change} = 2^{-\Delta\Delta CT} \quad (1)$$

where,

$$2^{-\Delta\Delta CT} = \frac{[(CT \text{ gene of interest} - CT \text{ internal control})]_{\text{sample A}}}{-[(CT \text{ gene of interest} - CT \text{ internal control})]_{\text{sample B}}} \quad (2)$$

Statistical analysis

The experiments were performed (*stx1* and *stx2* expression analysis) in duplicate, and results were expressed as means standard deviations for the repeated experiments as mean fold change. Mean fold change was converted to Log2 for symmetrical graphical representation. The results were analyzed using unpaired Student's *t* test (SPSS). $P=0.05$ were considered significant.

Results

Susceptibilities of isolates against antibiotics

Antibiotic susceptibility data using disk diffusion assay (Table 2) revealed that most of the isolates were resistant to all major antimicrobials tested including penicillin (aminopenicillins), fluoroquinolones, aminoglycosides, tetracycline, macrolides, carbapenems, and miscellaneous groups.

Effects of Sub MIC antibiotic concentrations on induction of Shiga toxin expression

Upon induction by selected antibiotics, the gene expression profiles of Shiga toxins were determined using qPCR. The data revealed considerable variations in stable mRNA levels of Shiga toxin gene relative to the levels of an internal control with and without induction by selected antibiotics. Results were calculated as fold change and expressed as log2 fold change. Ciprofloxacin substantially increased the expression of *stx1* and *stx2* genes to a level of 6.5 log2 (Fig. 1) and was found to be statistically significant ($P=0.01$). Fosfomycin acted as weak inducer for both shiga toxin 1 and 2 genes but mainly for *stx2*. Average expression weakly increased up to a significant level. The expression of *stx1* and *stx2* genes was recorded upto twofold (Fig. 2). Tigecycline also increased the expression level of *stx1* and *stx2* genes to a significant level of upto eightfold with a statistical significance of $P=0.001$ (Fig. 3). Meropenem, however, did not increase the expression level of *stx1* and *stx2* genes to a comparable level like other antibiotics. The expression of the genes rose upto 2-fold and was found to be statistically significant ($P=0.05$) after 24 h of incubation with Sub MIC of meropenem in comparison to culture without antibiotic (Fig. 4).

Resistance and fold gene expression of Shiga toxins

The expression of both the Shiga toxins 1 and 2 was higher in resistant strains than the sensitive ones upon induction by antibiotics. Figure 5 explains the expression profile of both Shiga toxin 1 (5A) and Shiga toxin 2 (5B) in resistant and susceptible strains.

Discussion

Shiga toxin-producing *Escherichia coli* (STEC) O157:H7 is a common foodborne pathogen associated with bloody diarrhea that may lead to hemolytic-uremic syndrome (HUS) if not treated properly. The release of shiga toxin (Stx) is the main reason for establishment and progress of Shiga toxigenic *Escherichia coli* (STEC) infections. The potential production of Stx 1 and Stx 2 from dead and live bacteria and the risk of developing antibiotic resistance have made the usage of antibiotics against STEC a controversial issue (Cheng et al. 2017). The effects of antibiotics on Stx production in the local isolates were investigated against empirically used antibiotics for their effects on induction of Stx-virulent pathogens. Expression of the shiga toxin genes was assessed through qPCR.

Antibiotic susceptibility assay of different classes of antibiotics was performed to determine the antibiotic susceptibility profile of the sampled isolates. The susceptibility analysis showed most of the isolates to be multidrug resistant. Highest resistance (66–87%) was found against penicillins, fluoroquinolones, and other higher generation antibiotic classes. MIC assay was performed since it provides a quantitative information for resistance as compared to simple classification by disk diffusion into susceptible, intermediate, or resistant phenotypes. Broth Microdilution assay was performed for MIC determination to test the four clinically used antibiotics (Reller et al. 2009; Wiegand et al. 2008). In order to study the relative gene expression of *stx1* and *stx2* genes, qPCR was performed, qPCR data was collected as cycle threshold values (Ct), and fold changes were calculated through comparative method (Eq. 1).

All of the four antibiotics showed varied results on Shiga toxin 1 and 2 expression. Previous reports from other geographical locations involving Shiga toxigenic *E. coli* O157:H7 had indicated that antibiotics like fluoroquinolones, trimethoprim/sulfamethoxazole, and ciprofloxacin significantly increased Stx 2 production (Grif et al. 1998; Kimmitt et al. 2000; Ochoa et al. 2007), whereas macrolides (Bielaszewska et al. 2012), carbapenems (Lee and Stein 2009), aminoglycosides (McGannon et al. 2010), and fosfomycin (Zhang et al. 2009) either had no effect on Stx2 production or suppressed it.

In our dataset, we observed that ciprofloxacin strongly increased both the shiga toxins mainly Shiga toxin 2 like previous reports (Botelho-Nevers et al. 2012). The expression of both the genes was upregulated with *stx2* on higher side. Ciprofloxacin inhibits bacterial growth by damaging DNA which triggers the activation of DNA repair mechanism, i.e., SOS response. Such activation leads bacteria to enter into the lysogenic phase of growth circle (Guerin et al. 2009). In present study, expression of the Shiga toxins was substantially increased to almost eightfold after treatment with ciprofloxacin.

Table 2 Average inhibitory effect of antibiotics and their susceptibility based on EUCAST breakpoint tables for interpretation of zone diameters. Version 7.0, 2017; S*indicates susceptibility; R* indicates resistance

	Penicillins		Fluoroquinolones		Cephalosporins		Carbapenems		Aminoglycosides	Tetracycline	Macrolides	Miscellaneous		
	S ≥ 14	S ≥ 19	S ≥ 22	S ≥ 25	S ≥ 27	S ≥ 14	S ≥ 25	S ≥ 19	S ≥ 22	S ≥ 17	S ≥ 18	S ≥ 14	S ≥ 17	S ≥ 24
	R < 14	R < 19	R < 22	R < 24	R < 21	R < 14	R < 22	R < 19	R < 16	R < 14	R < 15	R < 11	R < 17	R < 24
	SAM	AMC	MxP	CIP	FEP	KZ	CRO	FOX	MEM	IPM	CN	TGC	SxT	FOS
1	S	S	S	R	S	R	S	S	S	R	S	S	R	R
2	R	R	R	R	R	R	R	R	R	R	R	R	R	R
3	R	R	R	R	R	R	R	R	R	R	R	R	R	R
4	R	R	R	R	R	R	R	R	R	S	S	R	R	S
5	R	R	R	R	R	R	R	R	S	S	S	R	R	R
6	R	R	R	R	S	R	R	R	R	R	R	R	S	S
7	R	R	R	R	R	R	R	R	S	S	R	R	R	R
8	R	R	R	R	R	R	R	S	S	S	S	R	S	R
9	R	R	R	R	R	R	R	R	R	R	S	R	R	R
10	R	R	R	R	R	R	R	R	S	S	S	R	R	R
11	R	R	R	S	R	R	S	R	R	R	S	S	R	S
12	R	R	R	R	R	R	R	S	S	R	R	R	R	R
13	R	R	R	R	S	R	R	S	R	S	S	R	S	S
14	R	R	R	R	R	R	R	R	R	R	S	R	S	S
15	R	R	R	R	R	R	R	S	S	R	S	R	S	R

*The breakpoints were taken from CLSI document M100—S23 (CLSI 2013)

Fig. 1 Increase in *stx1* and *stx2* genes expression after treatments with Sub MIC of ciprofloxacin. Experiments were performed in duplicates and are shown as mean fold change, that was converted to Log2 for symmetrical graphical representation

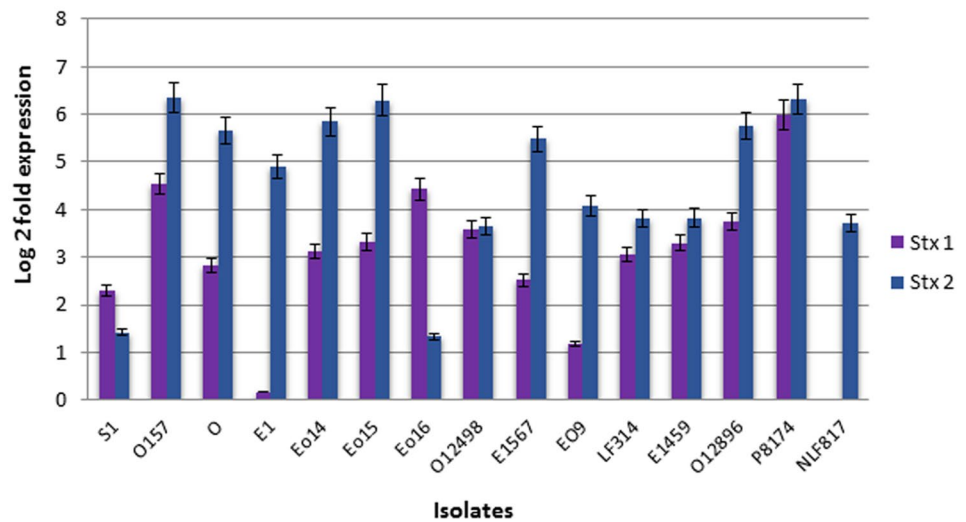


Fig. 2 Change in the expression level of *stx1* and *stx2* genes induced by fosfomycin. The antibiotic is a weak inducer of both the genes

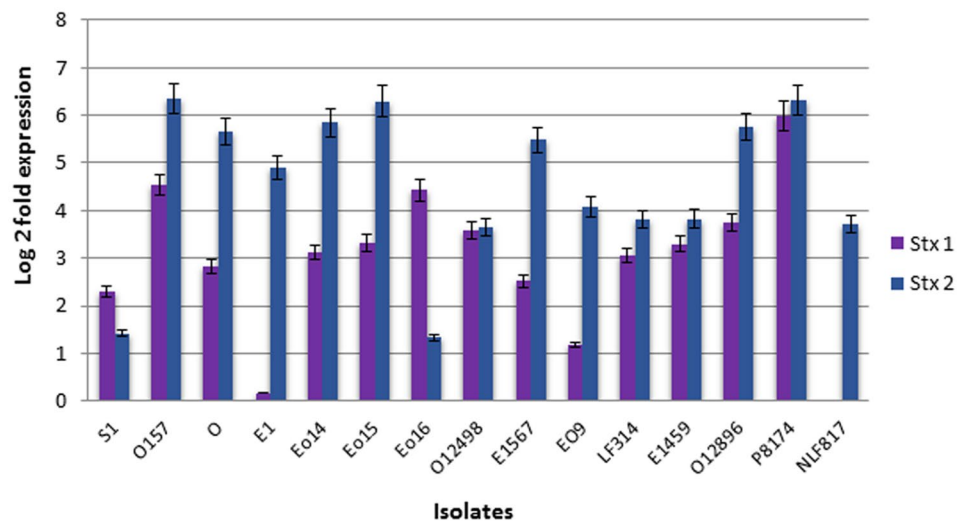


Fig. 3 Change in the expression level of *stx1* and *stx2* genes induced by tigecycline. The drug showed a significant induction of Shiga toxin genes

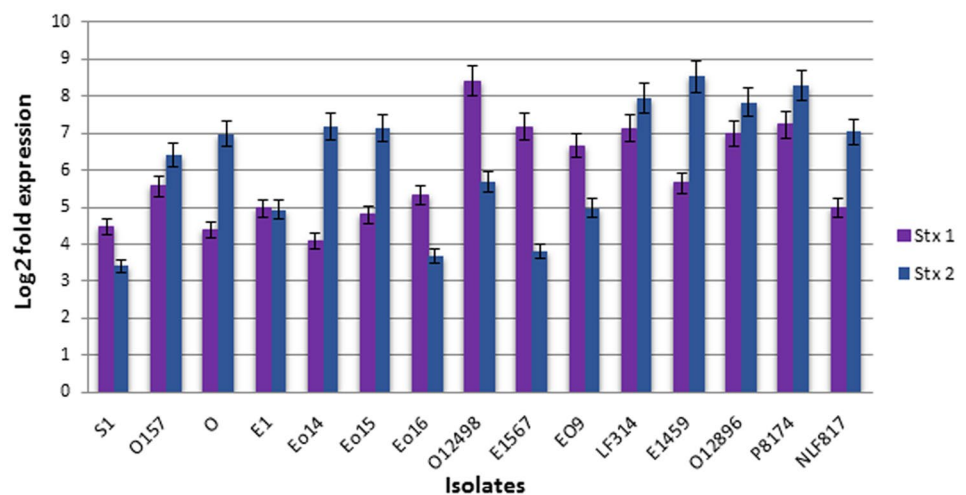
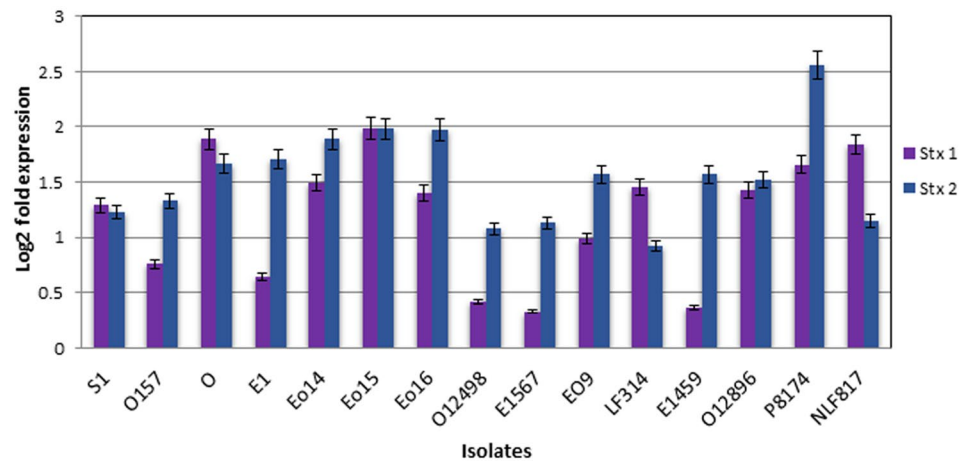


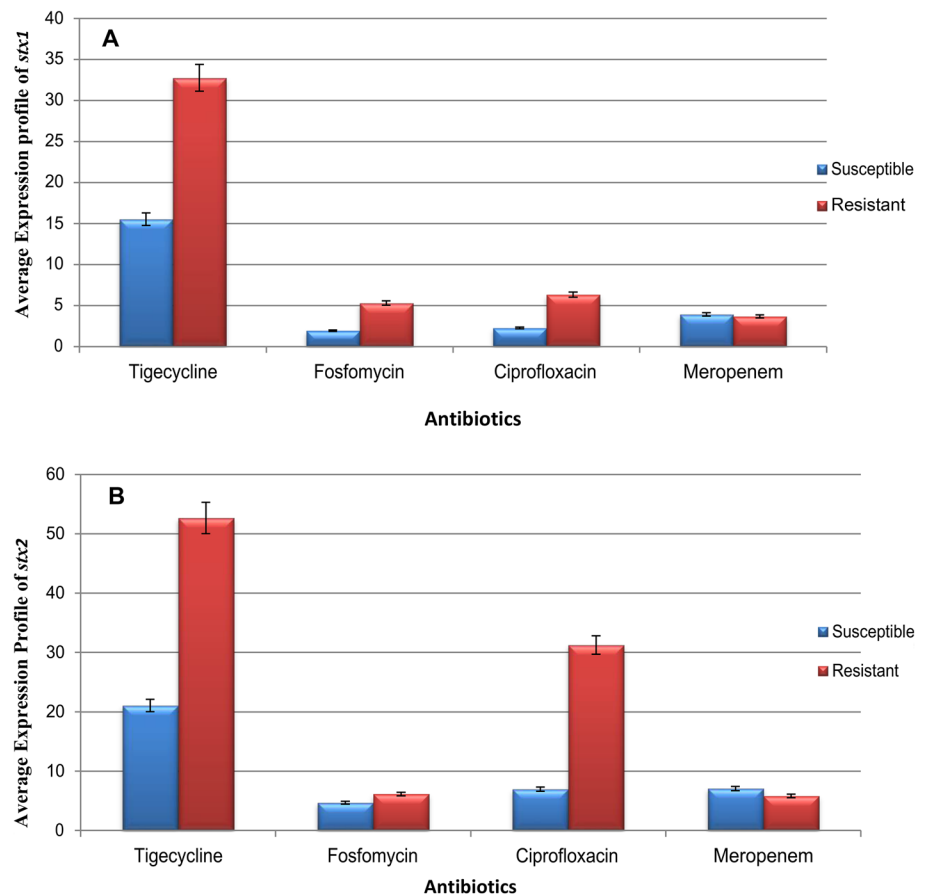
Fig. 4 Change in the expression level of *stx1* and *stx2* genes induced by meropenem. The drug is the weak inducer than other antibiotics



Fosfomycin [(1,2-epoxipropyl) phosphonic acid] is an antimicrobial agent that damages the bacterial cell wall and has previously been linked with increased production of shiga toxins. Fosfomycin inhibits the formation of N-acetylmuramic acid, a major component of bacterial cell walls. It has bactericidal activity against actively growing, susceptible organisms (Arca et al. 1988). As it is not involved in

DNA damage, it does not affect SOS pathway attributing to weak induction on shiga toxin expression in Sub MIC dose (McGannon et al. 2010). Effect of the antibiotic is dose dependent, and it also varies among isolates from different geographical distribution. In the present study, fosfomycin weakly induced shiga toxin 1 and 2 expressions barely by twofold.

Fig. 5 Expression profile of **A** *stx1* and **B** *stx2* in resistant and susceptible strains against selected antibiotics. The expression of Shiga toxins genes was higher in resistant strains as compared to sensitive strains that explains the effect of induction by antibiotics



Tigecycline belongs to a class of tetracycline, which inhibit protein synthesis by interrupting binding of incoming aminoacyl transfer RNAs to the ribosomal acceptor site, thus halting the elongation step in protein synthesis (Hancock 2005). Bielaszewska et al. (2012) reported that tigecycline could not induce shiga toxin production in an outbreak strain. Though most of the isolates were susceptible to tigecycline in our study, even then it could strongly increase the expression of both shiga toxins to a level of eightfold.

Meropenem is a member of carbapenems β -lactam antibiotics which target penicillin binding proteins located in the bacterial cell wall. Being a relatively newer antibiotic, resistance to meropenem was found to be intermediate in all isolates, consistent with the previous reports (Ibarra et al. 2013). Our results showed that the shiga toxins were not affected by meropenem to a significant level.

These data of expression analysis may help in deducing a probable relationship between resistance and pathogenesis. The resistant isolates generally exhibited higher changes in toxin expression regardless of shiga toxin 1 and 2. In case of *E. coli* and shiga toxin production, a DNA repair mechanism, namely, SOS response has been reported to be directly linked with both resistance and production of toxins (Nassar et al. 2013). Although the complete mechanism of interaction between resistance and pathogenesis is yet to be discovered, the possible interlink between the two phenomenon may be explained by the fact that sequences of Shiga toxin are reportedly encoded by a lambdoid bacteriophage, which may halt host cell protein synthesis to establish the infection. Some of the Phage particles carrying antibiotic resistance genes (ARGs) have also been described in a range of environments (Gómez-Gómez et al. 2019).

Conclusion

In general, ciprofloxacin and tigecycline were significantly involved in up regulation of both the Shiga toxins. Although both the genes were up regulated, *stx2* was found to be more affected by Sub MICs of ciprofloxacin and tigecycline. Fosfomycin weakly induced the expression of *stx2* only, whereas meropenam was the least inducer. Meropenem could be an effective option for treating shiga toxigenic *E. coli* local isolates in Pakistan.

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Author contribution All authors contributed to the study conception and design. Material preparation, data collection, and analysis were

performed by Anika Rehman, Saadia Andleeb, Sidra Rahmat Ullah, and Khalid Mehmood. The first draft of the manuscript was written by Anika Rehman, Saadia Andleeb, Sidra Rahmat Ullah, Zeeshan Mustafa, and Danish Gul. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

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